

ABORTION

By

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ABORTION

Definition

- Termination of pregnancy **before viability**.

Ethiology

1. Chromosomal abnormalities:
 - Cause at least 50% of early abortions e.g. trisomy, monosomy X & triploidy.
2. Blighted ovum (anembryonic GS): where there is no visible fetal tissue in the sac.
3. Maternal infections

4. Trauma :

- External to the abdomen or during abdominal or pelvic **operations**.

5. Endocrine causes:

- Progesterone deficiency
- DM
- Hyperthyroidism

6. Drugs & environmental causes:

- e.g. Quinine, ergots, severe purgatives, tobacco, alcohol, arsenic, lead, formaldehyde, benzene & radiation

7. Maternal anoxia & malnutrition.

8. Overdistention of the uterus.

9. Immunological causes:

- SLE
- Antiphospholipid antibodies that are directed against PLTs & vascular endothelium leading to **thrombosis, placental destruction** & abortion.
- **Histocompatibility** between the mother & father and in turn the fetus.
 - It is assumed that histocompatibility particularly in human leukocyte antigen (**HLA- DR locus**) is essential for stimulation of the immune system to produce blocking factors which prevent **rejection of the fetus**.

10. Aging sperm or ovum.

11. Psychological conditions & over fatigue.

12. Uterine defect

- Septum, Asherman's syndrom,
Subcutaneous myoma.
- Cervical incompetence

Cervical incompetence

- Clinically result in **midtrimester** spontaneous miscarriage or **early preterm delivery**.
- The miscarriage tends to be **rapid, painless & bloodless**.
- Dx :
 - By passage of a **Hegar 8 dilator** without difficulty in non pregnant woman or by U/S exam. Or **premenstrual hysteroqram**.

Classification

By occurrence

1. Spontaneous abortion
2. Induced abortion
 - Safe
 - Unsafe
3. Therapeutic abortion

Classification

By clinical stages

- A. Threatened Abortion
- B. Inevitable Abortion
- C. Incomplete Abortion
- D. Complete Abortion
- E. Missed Abortion

Threatened abortion

- Vaginal bleeding
- Uterus is enlarged
- **Cervix is closed**
- Lower abdominal pain is either minimal or absent
- The fetus is alive & there is a chance of continuing the pregnancy to viability.

Inevitable abortion

- Vaginal bleeding of variable amount.
- Crampy lower abdominal pain.
- **Cervix is open.**
- No products of conception have been expelled.
- There is no chance of salvaging the pregnancy.

Incomplete abortion

- Vaginal bleeding.
- Cervix is open.
- Expulsion part of the products of conception.

Complete abortion

- Vaginal bleeding stops.
- Pain will cease.
- Cervix closes following expulsion of products of conception.

Missed abortion

- Retention of the fetus in utero after death.
- There is usually Hx of threatened abortion preceding it.
- Brownish discharge.
- Pregnancy symptoms like morning sickness, breast tenderness & abdominal increment disappear.
- Cessation of fetal movement is reported by the mother if it occurs after 18 wks.
- Failure of uterine growth small for GA uterus.



Septic abortion

- Offensive vaginal discharge.
 - Fever
 - Lower abdominal pain/ tenderness
- +
- Any of the clinical stages of abortion.

Diagnosis

Initial assessment

- Suspect abortion in any woman of reproductive age experiencing at least 2 of the following symptoms.
 - Vaginal bleeding.
 - Crampy & lower abdominal pain.
 - Amenorrhea

Diagnosis

Hx

- Length of amenorrhea
- Bleeding (duration, amount)
- Cramp (duration, severity)
- Abdominal or shoulder pain.
- Drug allergy.
- Hx of interference & method employed.
- Symptoms of infection.

Diagnosis

Physical examination

- V/S
- Note general health of the woman.
- General systemic exam.
- Abdominal exam.
 - **Check:** abd. Distention, mov't with respiration, bowel sounds, location & severity of tenderness, rebound tenderness, uterine size, mass, shifting dullness.

Diagnosis

Pelvic examination

- Remove any visible POC from the vaginal canal or cervical canal.
- Note the amount of bleeding.
- Presence of offensive discharge.
- Extent of cervical dilatation.
- CET
- Size & consistency of the uterus, adnexal mass & other pelvic masses.
- Check for cervical laceration.

Diagnosis

Lab. Examination

Based on clinical assessment when indicated:

- Hg/ Hct, BG & Rh-factor
- WBC, ESR, U/A
- RFT, LFT, PLT, PT, PTT
- Plain film of abdomen (erect)
- U/S
- Pregnancy test.

Management

- Life threatening conditions like shock severe anemia & sepsis should be treated aggressively prior to instituting specific treatment.

Management

Threatened abortion

- Expectant management, reduce activity and avoid sexual intercourse

Management

Complete abortion

- If completeness is justified either by examination of conceptus tissue or by u/s
 - Ergometrine 0.5 mg im
 - If justified provide therapeutic or prophylactic antibiotics.

Management

Inevitable abortion

1. GA < 14 wks

- Evacuation

2. GA > 14 wks

- Evacuation not advised before expulsion of the fetus.
- Admission & monitoring the V/S & amount of bleeding.
- Once the fetus/ placenta are expelled completeness should be checked.
- Evacuation of the uterus must be done if incomplete or bleeding continues.

Management

Incomplete abortion

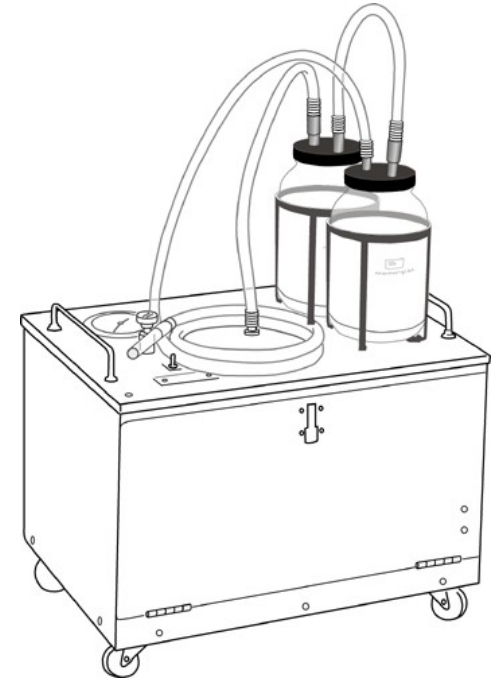
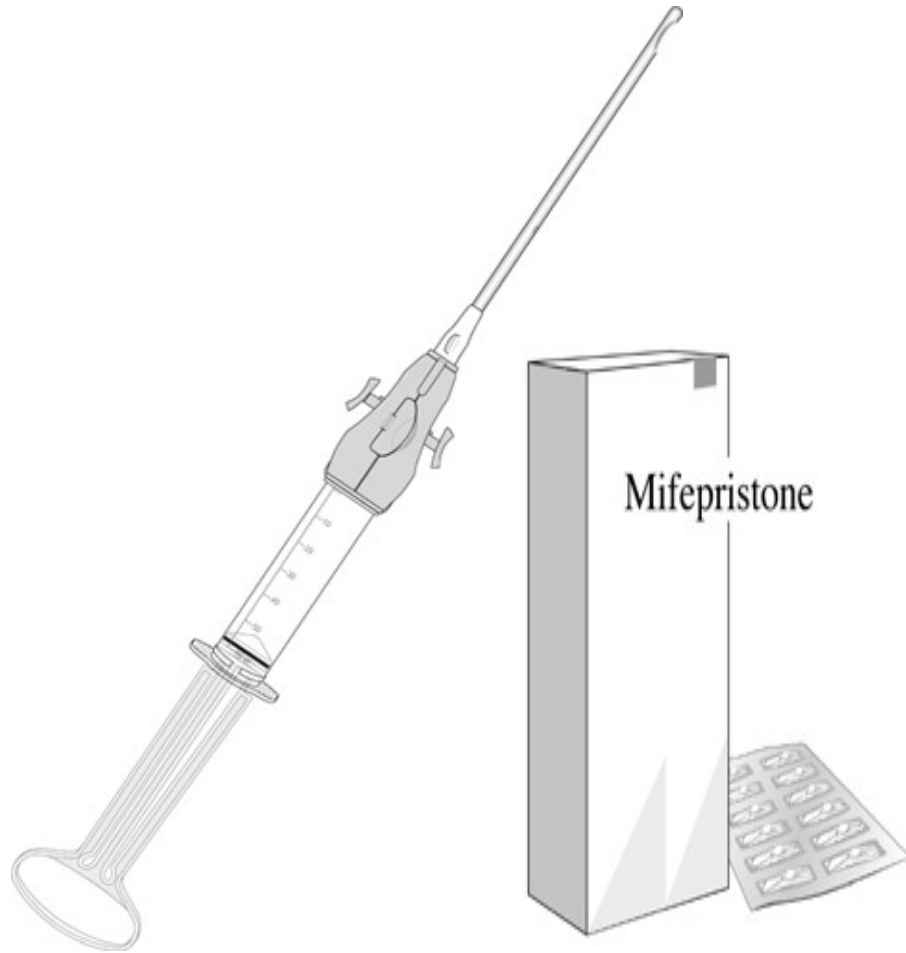
- Evacuation

Management

Missed abortion

- Evacuation

Uterine Evacuation Methods



Uterine Evacuation Methods

- Uterine evacuation removes contents of the uterus
- Several methods for actively managing uterine evacuation
 - Vacuum aspiration techniques
 - Surgical techniques
 - Medication or pharmacological techniques

Expectant Management

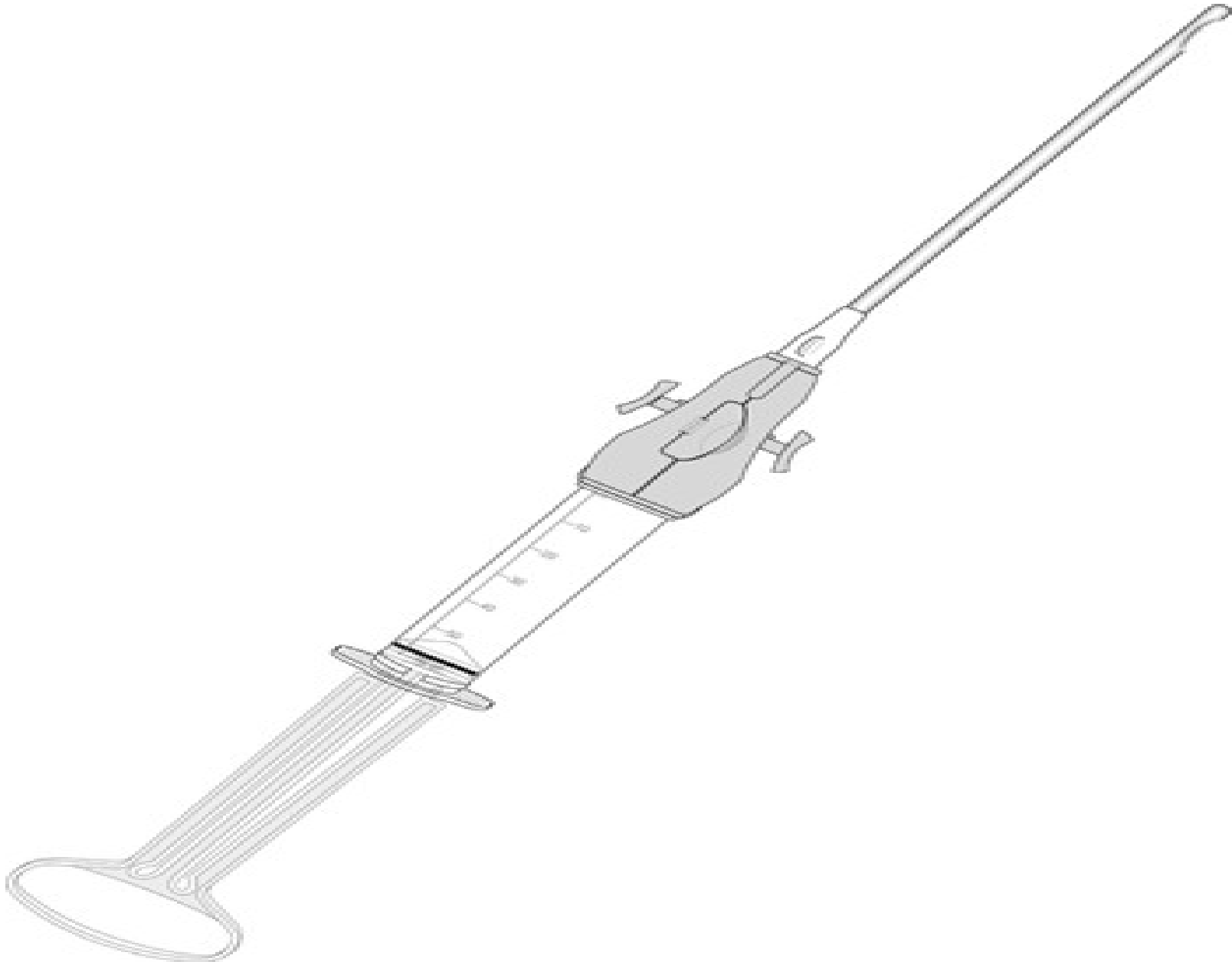
- Used in cases of **inevitable abortion**
- Spontaneous abortion may completely expel products
- No intervention: allow natural expulsion to occur
- Providers need to closely monitor the woman
- Need to be able to quickly access emergency care

Which Method is Best?

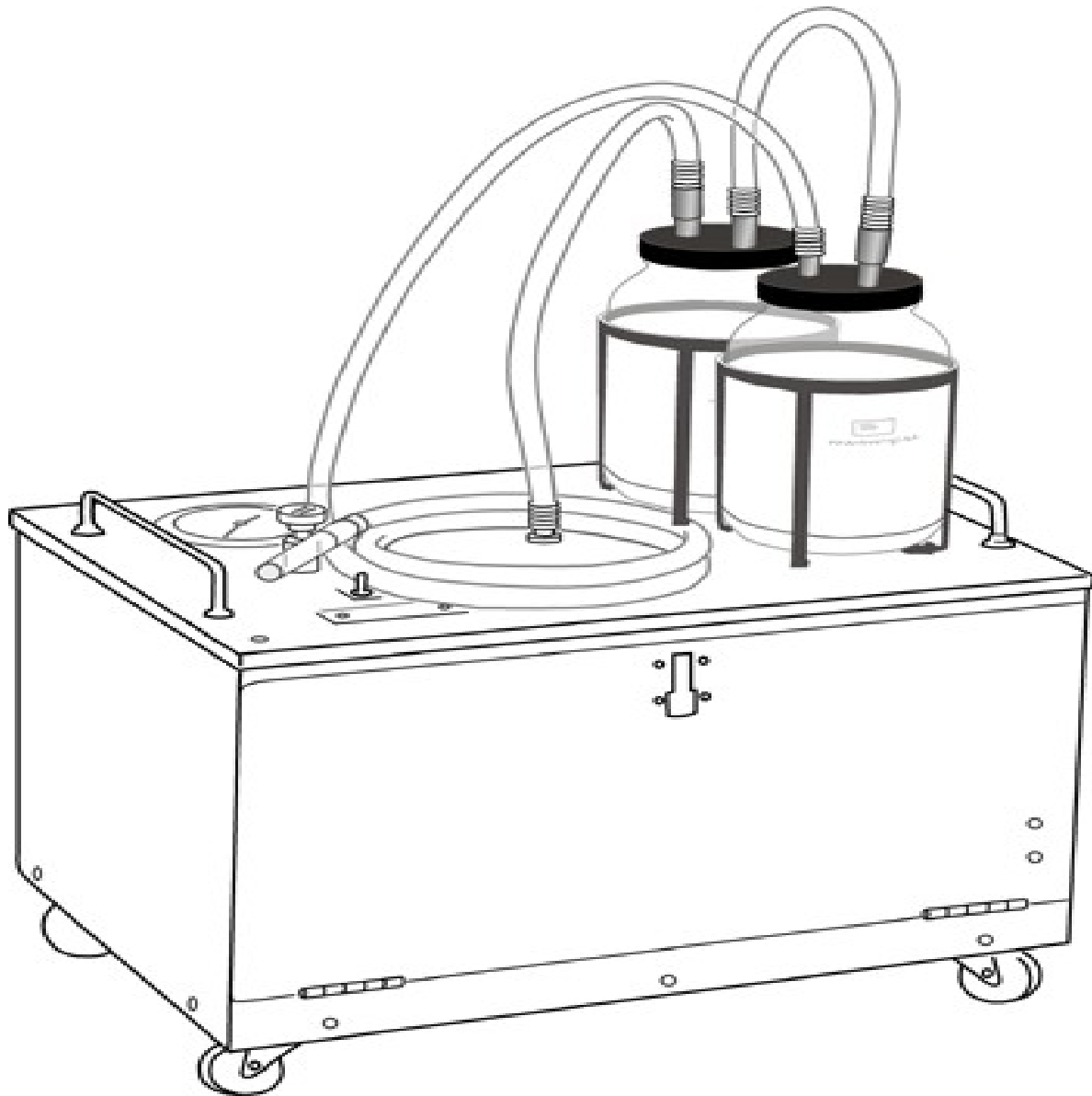
Use of a certain method depends on:

- Staff skills
- Equipment, supplies, drugs available
- The woman's clinical condition
- Her personal preference

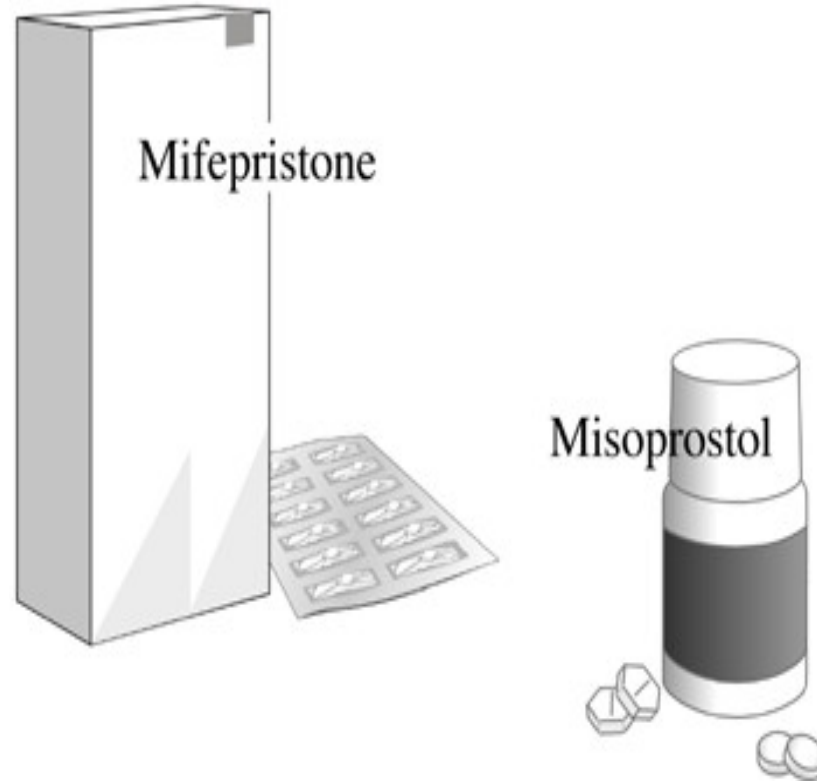
Manual Vacuum Aspiration (MVA)



Electric Vacuum Aspiration (EVA)



Medication or Pharmacological Methods



According to the World Health Organization (WHO)

- Vacuum aspiration (VA) is preferred method for PAC in the first trimester
- WHO recommends replacing SC (sharp curettage) with VA

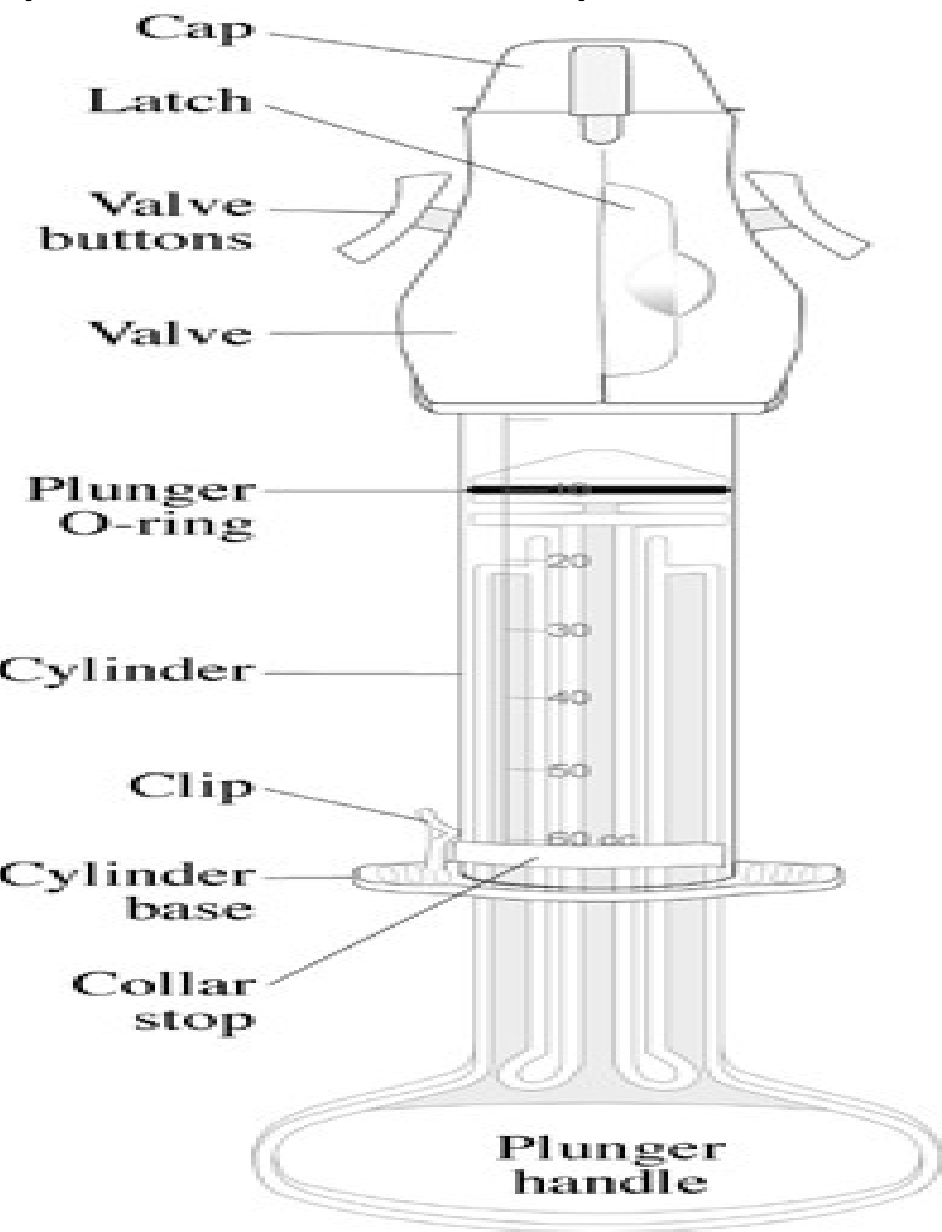
According to the International Federation of Gynecology and Obstetrics and other organizations

- Properly equipped hospitals should select manual vacuum and/or electric vacuum according to the expertise available.” (FIGO/WHO Taskforce, 1997)
- MVA is an essential technology for uterine evacuation. (UNFPA, ICM, UNICEF, World Bank)

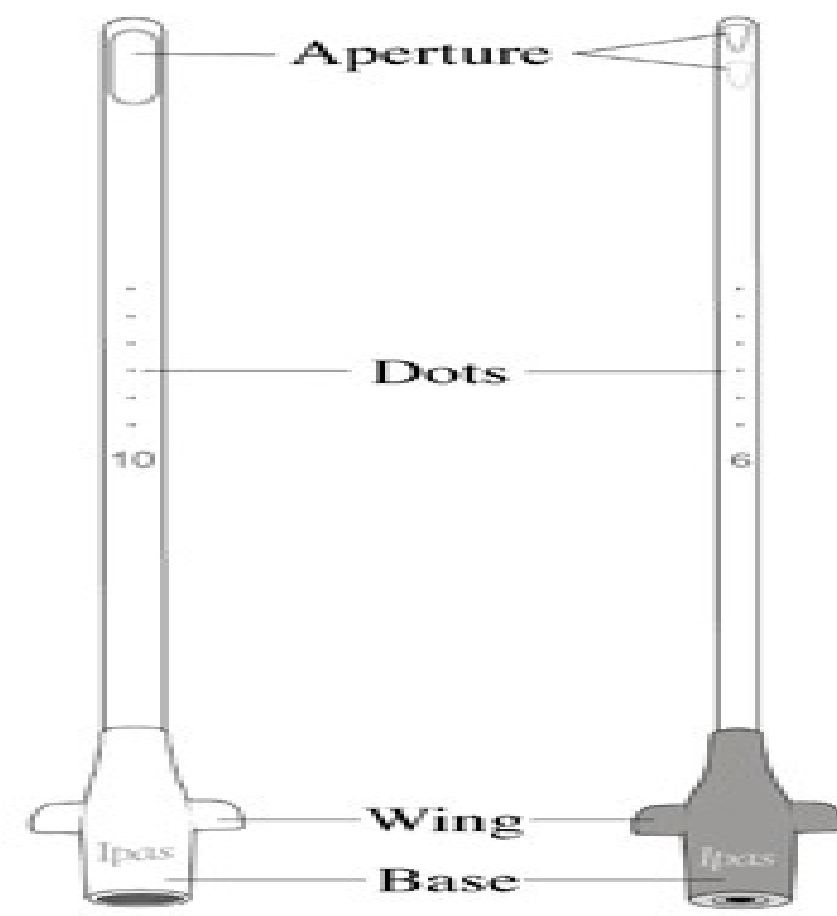
Benefits of VA as compared to sharp curettage

- Reduced risk of infection
- Reduced risk of cervical injuries & uterine perforation
- Reduced need for cervical dilatation
- Blood loss is less
- Shortened hospital stay
- Decreased need for anesthetic drugs

Ipas MVA Plus™ Aspirator and Ipas EasyGrip® Cannulae

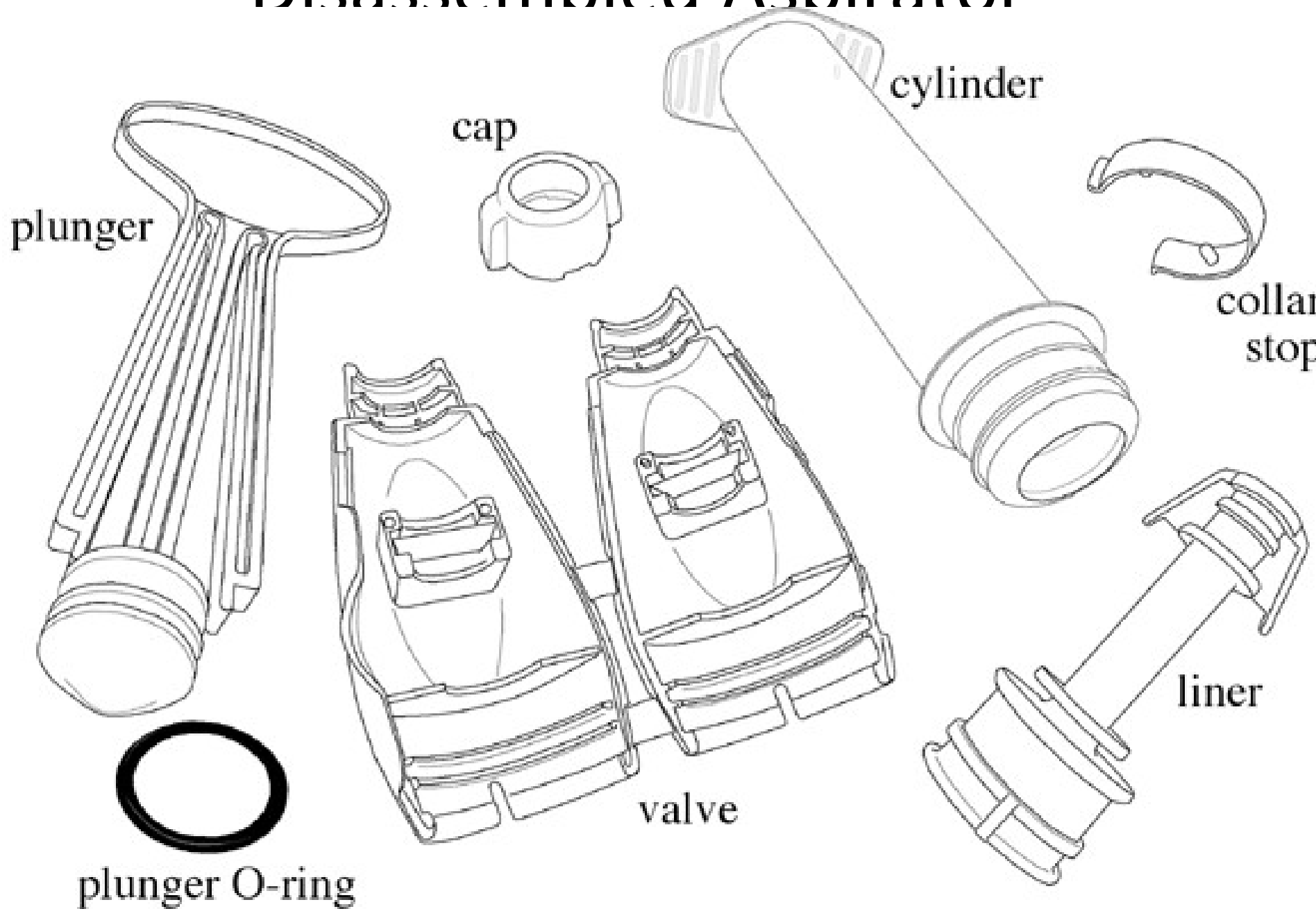


Ipas MVA Plus™



Ipas EasyGrip® Cannulae

Disassembled Aspirator



Ipas MVA Plus™ Aspirator and Ipas EasyGrip® Cannulae

Intended use and indications:

- Intended for uterine aspiration or evacuation
- Treatment of incomplete abortion for uterine size up to 12 weeks LMP
- First-trimester abortion (menstrual regulation)

Ipas MVA Plus™ Aspirator and Ipas EasyGrip® Cannulae (cont.)

- Endometrial biopsy
- Only contraindication: endometrial biopsy
in cases of **suspected pregnancy**

Selection of Cannulae

- Appropriate to **uterine size** and amount of **dilation**
- Uterine size 4–6 weeks LMP: suggest **4–7mm**
- Uterine size 7–9 weeks LMP: suggest **5–10mm**
- Uterine size 9–12 weeks LMP: suggest **8–12mm**

Steps of the MVA Procedure

1. Prepare instruments
2. Prepare the woman
3. Perform cervical antiseptic prep
4. Administer paracervical block
5. Dilate cervix
6. Insert cannula

Steps of the MVA Procedure

7. Suction uterine contents
8. Inspect tissue
9. Perform any concurrent procedures
10. Process instruments

Step 1: Prepare Instruments

- Check vacuum retention of aspirator
- Have more than one aspirator available

Step 2: Prepare the Woman

- Ensure pain medication is given at appropriate time
- Ask the woman to **empty her bladder**
- Help her onto the table
- Ask for her **permission** to start
- Put on barriers and wash hands
- Perform **a bimanual exam**

Wear Barriers for MVA Procedures

Wear gloves when you expect to get blood or body fluids on your hands.

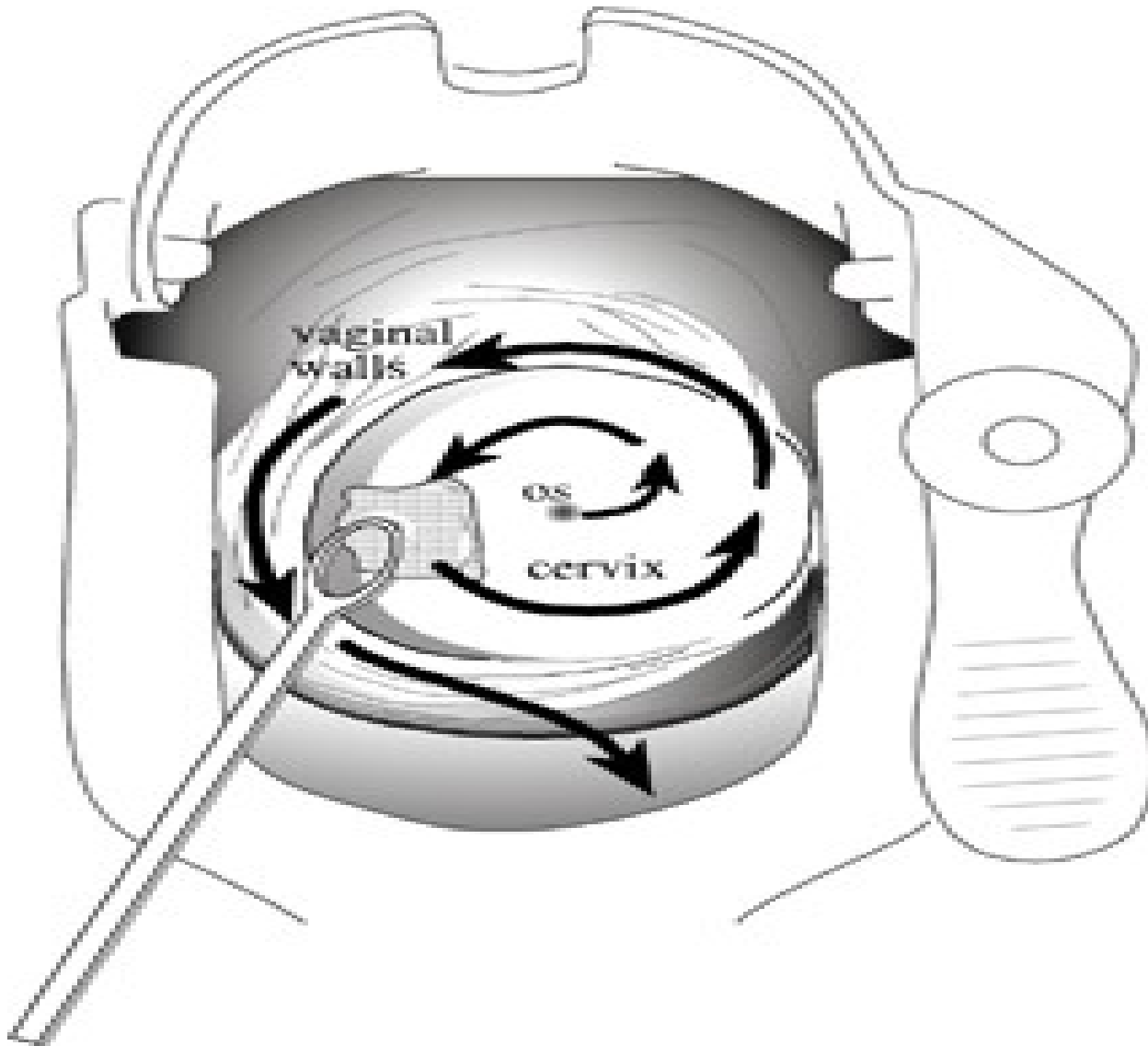


Add face protection when you may have blood or fluids splashed.

Step 3: Perform Cervical Antiseptic Prep

- Follow No-Touch Technique
- Use antiseptic sponges to **clean cervix and os** and then, if desired, vaginal walls
- Do not retrace areas previously cleaned

Cervical Antiseptic Preparation



Step 4:

Administer Paracervical Block

- If os is open, may not be necessary
- Recommended for all uterine evacuation patients
- Injection sites vary but technique accepted globally

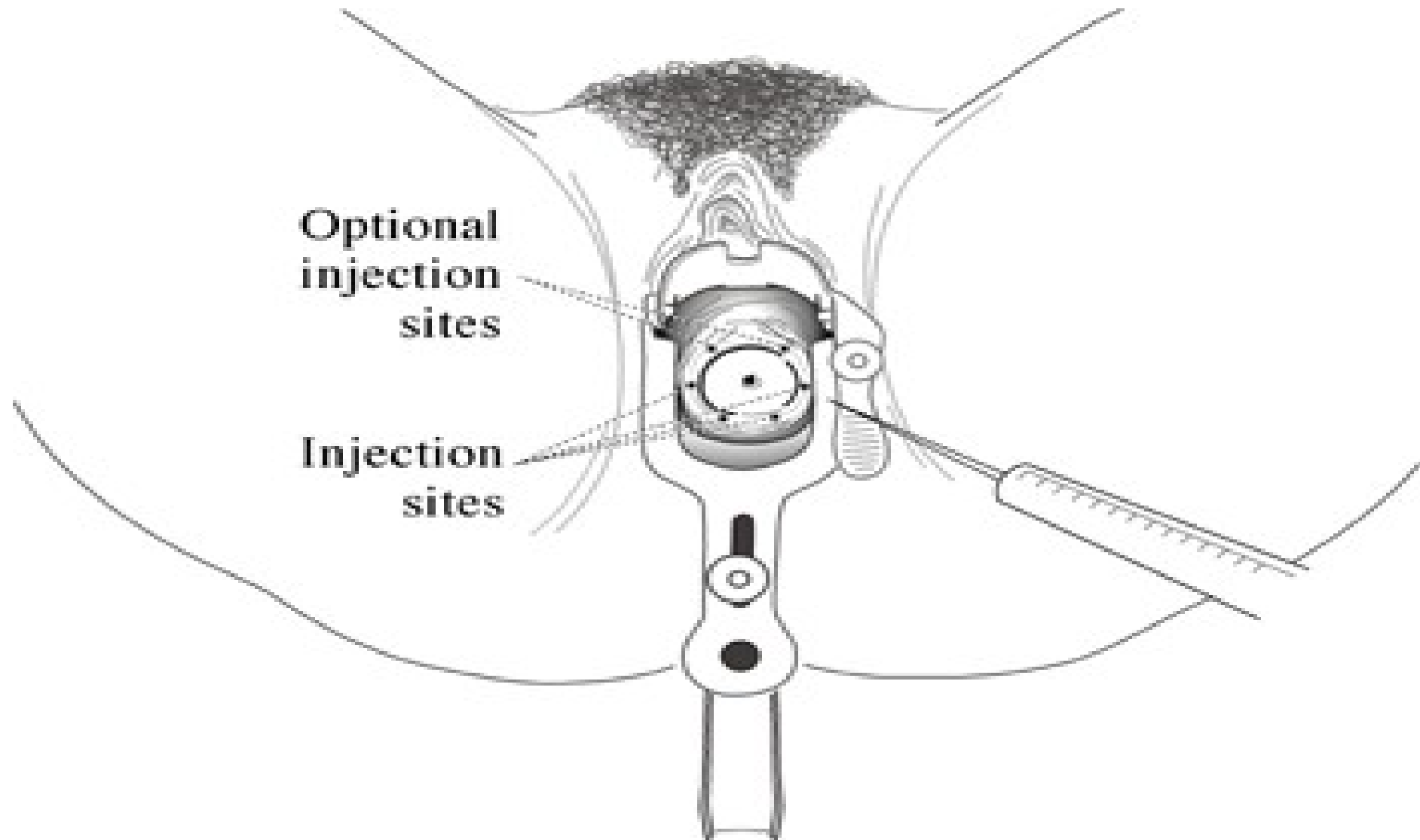
Step 4: Administer Paracervical Block (cont.)

- Usually **10-20mL** of 0.5%-1.0% lidocaine (always less than 200mg)
- Always aspirate with needle before injecting

Administering Paracervical Block

- Inject 1-2mL of anesthetic where **tenaculum** will be placed
- Place tenaculum
- Apply **slight traction** to move cervix, exposing transition from cervical to vaginal tissue
- **Slowly** inject **2-5mL** of lidocaine into this tissue to depth of **1-1.5** inches at **3, 5, 7 and 9** o'clock

Paracervical Block



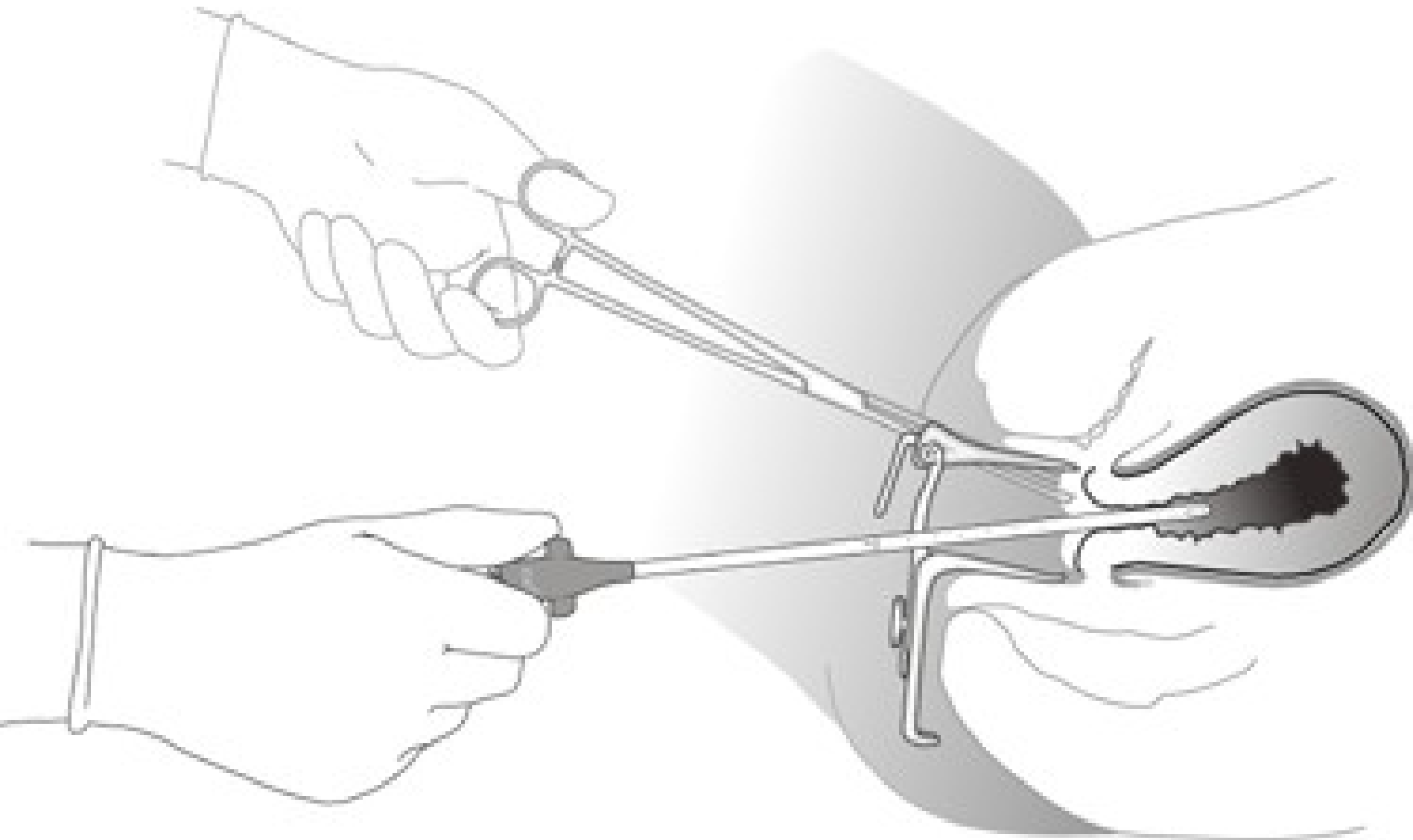
Step 5: Dilate Cervix

- Required in some but not all cases
- Cannula should fit snugly in os to hold vacuum
- Use gentle operative technique
- Use progressively larger series of cannulae or mechanical dilators
- Can use misoprostol

Step 6: Insert Cannula

- Gently apply traction to the cervix
- Rotate the cannula while gently applying pressure
- Insert cannula slowly until it touches the fundus, then draw back
- Alternatively, insert just past internal os

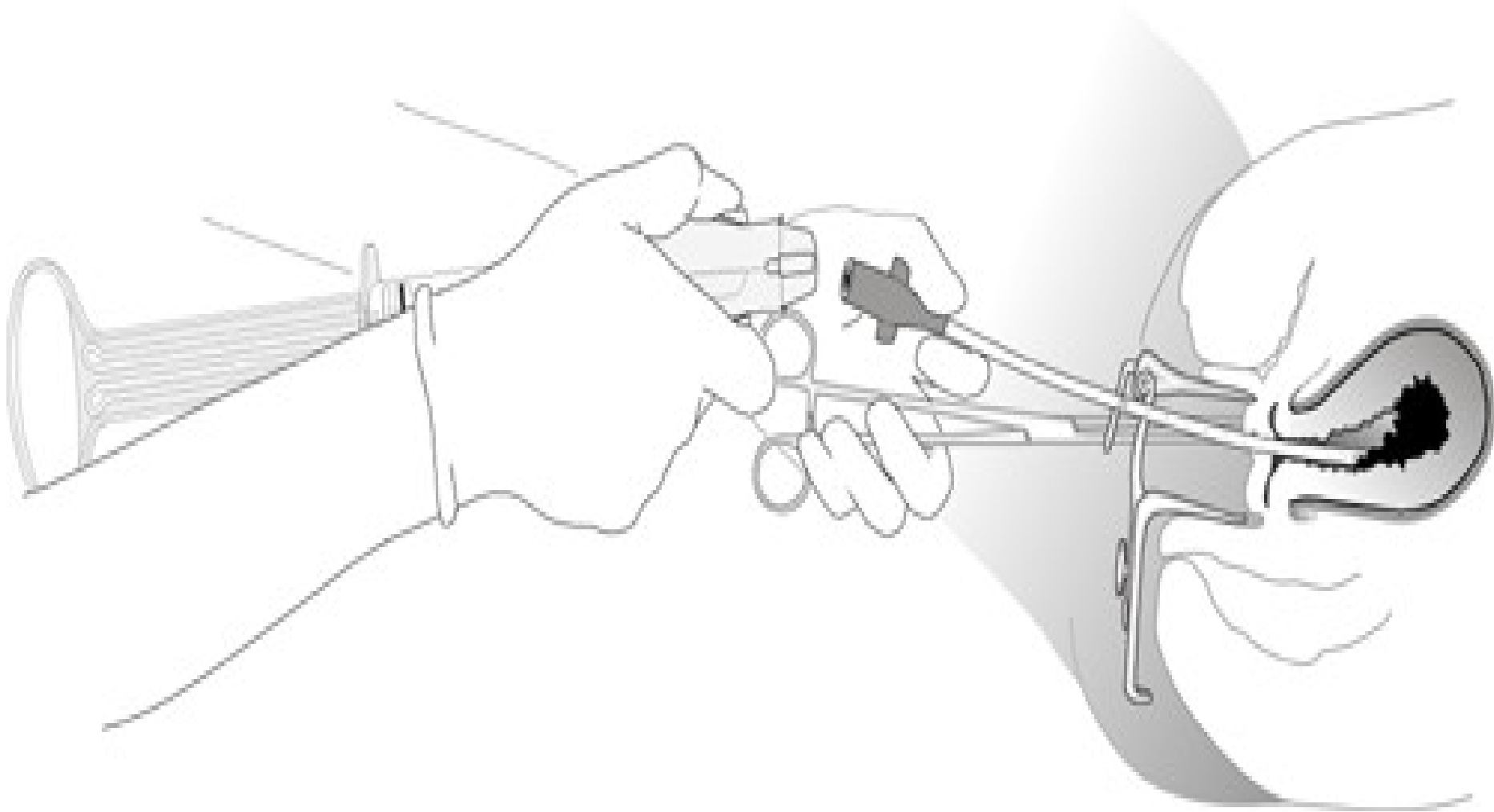
Insert Cannula



Step 7: Suction Uterine Contents

- Attach **charged aspirator**

Attach Aspirator

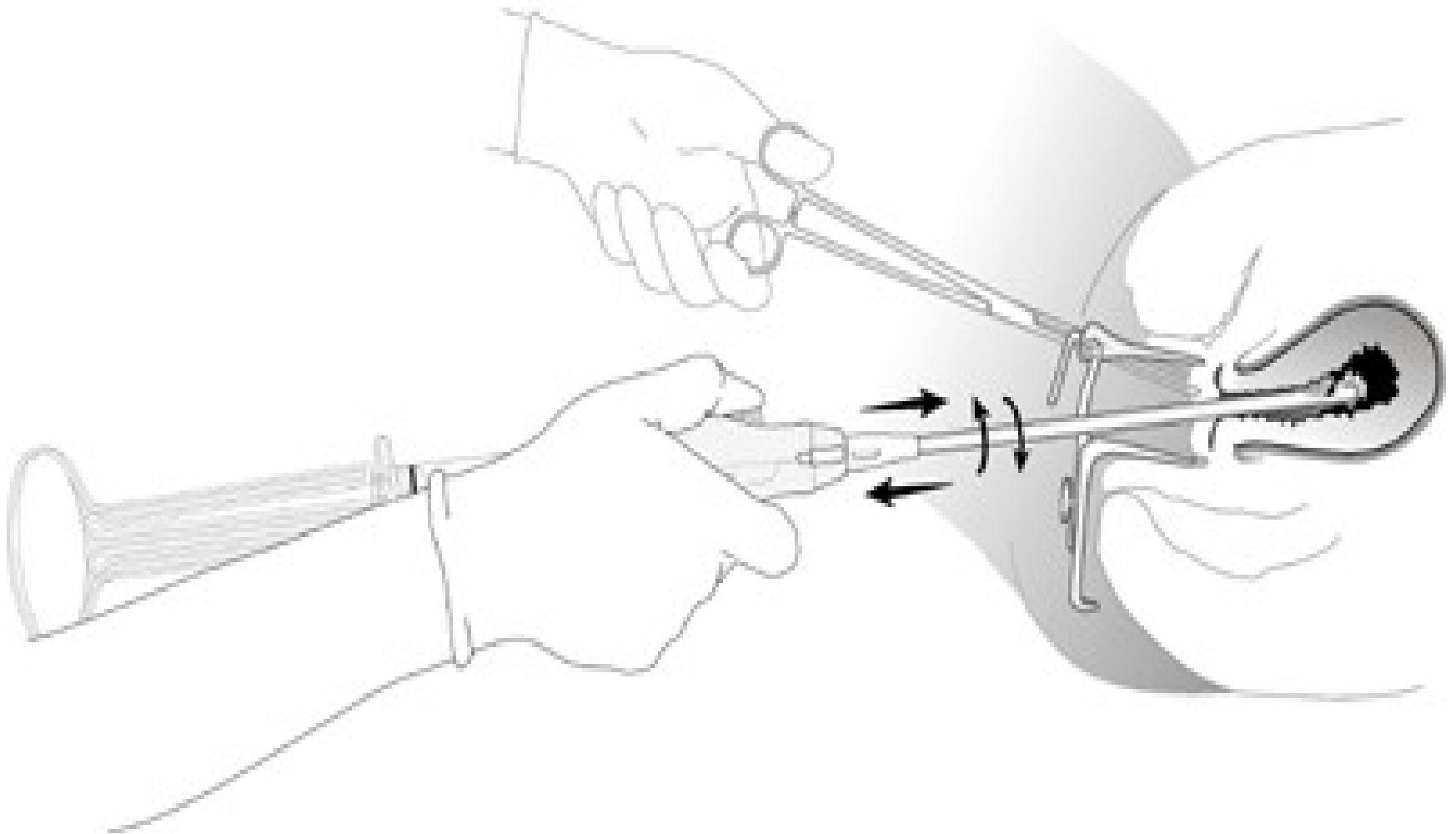


Step 7:

Suction Uterine Contents (cont.)

- Release buttons to start suction
- Gently rotate cannula 180 degrees in each direction
- Use “in and out” motion
- Do not withdraw cannula opening beyond external OS

Evacuate Uterine Contents



Signs That the Uterus is Empty

- Red or pink foam without tissue passing through cannula
- Gritty sensation over surface of uterus
- Uterus contracting around cannula
- Increased uterine cramping

When the Procedure is Finished

- Push **buttons down** and **forward** to close valve
- Disconnect cannula from aspirator OR
- **Remove cannula** from uterus without disconnecting
- May evacuate again after inspecting POC

Use Care to Disconnect Cannula

- Be aware that Ipas EasyGrip[®] cannulae fit firmly into the aspirator valve
- Use care when disconnecting cannula from aspirator

Step 8: Inspect Tissue

- Empty contents of aspirator into container
- Look for **POC: villi and decidua** should be visible

Inspect Tissue



Inspect POC

- Evaluate amount based on estimated gestation
- Determine that all POC have been removed
- If in doubt, strain POC, float in water or vinegar, view with light underneath

Abnormalities of POC

No POC visible? Possible reasons:

- Uterus emptied before procedure
- Uterine cavity still contains POC
- Ectopic pregnancy
- Misdiagnosis of pregnancy
- Uterine anatomical variations prevented evacuation

Abnormalities of POC (cont.)

Less than expected amount of POC? Possible reasons:

- Incomplete procedure: re-evacuate
- Gestational estimation incorrect

Step 9:
Perform Any Concurrent Procedures

If POC are satisfactory:

- Wipe the cervix with swab to assess additional bleeding
- Perform bimanual exam to check uterine size and firmness, if advised
- Perform concurrent procedures

Step 10:

Process Instruments

Process or discard instruments immediately

Post-Procedure

- Reassure the woman that the procedure is finished
- Help her into a comfortable position
- Ensure she is escorted to the recovery area

Decrease in MVA Vacuum

- Aspirator is full
- Cannula is withdrawn past os
- Cannula is clogged
- Aspirator is incorrectly assembled

When Aspirator is Full

1. Close the valve
2. Detach the cannula and leave in os
3. Open the valve and squeeze plunger arms
4. Push plunger and empty the aspirator
5. Establish new vacuum
6. Reattach to the cannula and continue

Use Care to Disconnect Cannula

- Be aware that Ipas EasyGrip® cannulae fit firmly into valve
- Care should be used when disconnecting cannula from aspirator

When Cannula is Withdrawn Past Os

1. Remove cannula and aspirator; don't touch vaginal walls
2. Detach and empty aspirator
3. Re-establish vacuum
4. Reconnect aspirator to cannula
5. Continue evacuation

When Cannula is Clogged

- Ease it back toward, not through, external os OR
- Close valve and withdraw cannula out of uterus
- Remove tissue clogging cannula using forceps
- Reinsert cannula and continue aspiration
- Never push tissue through cannula while in uterus

Loss of Vacuum

- Reassemble aspirator
- Check O-ring for damage

Complications

When Can Complications Develop?

- With induced or spontaneous abortion
- Before, during or after medical assistance
- Individually or several at the same time

Severe Complications

- Shock
- Severe vaginal bleeding
- Sepsis
- Intra-abdominal injury

Shock

Signs and Symptoms

- Anxious, restless
- Fast **heart rate** (over 110 beats per minute)
- **Fast, shallow respirations** (30 or more per minute)
- **Paleness** of mouth, fingertips

Shock (cont.)

Signs and Symptoms

- Systolic pressure less than 90mmHg
- Hemoglobin 8g/100mL or more
- Hematocrit more than 26%
- Clear lungs
- Cool, clammy skin, but she is sweating

Shock

Causes

- Severe blood loss
- Intra-abdominal injury
- Severe infection

Shock (cont.)

Initial Treatment

- Monitor vital signs and blood loss
- Oxygen at 6–8L/minute
- No fluids by mouth if IV fluids available or surgery needed
- Rapidly infuse IV fluids; refer if not available

Shock (cont.)

Initial Treatment

- Monitor intake and output
- Assess for blood transfusion
- Administer antibiotics
- Remove protruding POC

Severe Vaginal Bleeding

Signs and Symptoms

- Heavy, bright **bleeding**
- Paleness of mouth, fingertips
- Dizziness, fainting
- Significant changes in **vital signs**

Severe Vaginal Bleeding

Possible Causes:

- **Injury** to vagina, cervix, uterus, intra-abdominal organs
- Missed, inevitable or incomplete abortion
- Molar or ectopic pregnancy

Severe Vaginal Bleeding

Initial Treatment

- Monitor vital signs
- Ensure airway is open
- Evaluate blood loss
- Do not give anything by mouth
- Give oxygen at 6–8L/minute

Severe Vaginal Bleeding (cont.)

Initial Treatment

- Administer IV fluids
- Monitor intake, output
- Assess need for blood transfusion
- Administer antibiotics
- Medication for pain

Severe Vaginal Bleeding (cont.)

Initial Treatment

- Hemoglobin, cross-match, clotting tests
- Remove any protruding POC
- Refer if MVA and IV fluids not available
- Administer uterotonics; perform uterine massage

Sepsis

Signs and Symptoms

- Fever (38°C or more); chills, sweats
- Foul-smelling discharge
- Lower abdominal pain
- Prolonged bleeding
- Unsafe or spontaneous abortion
- Foreign body or IUD present

Sepsis

Contributing factors

- Retained POC (incomplete abortion)
- Injuries from unsafe abortion
- Infection prevention not followed
- Pre-existing infection; RTI/STD

Sepsis

Initial Treatment

- Monitor vital signs
- Ensure airway is open
- Do not give anything by mouth
- Oxygen at 6–8L/minute
- IV fluids; refer if unavailable

Sepsis (cont.)

Initial Treatment

- Monitor intake and output
- Administer broad-spectrum antibiotics
- Hemoglobin, cross-match, clotting tests

Intra-Abdominal Injury

Signs and Symptoms

- Abdominal pain, distension, shoulder pain
- Rebound; cervical motion tenderness
- Nausea; fever higher than 38°C
- Vaginal or internal bleeding, or both
- Decreased bowel sounds

Intra-Abdominal Injury

Common Causes

- Interference with pregnancy by unqualified persons or woman herself
- Proper procedure by qualified provider but with problems
- Ruptured ectopic pregnancy

Intra-Abdominal Injury

Initial Treatment

- Monitor vital signs
- Raise her legs
- Ensure airway is open
- Oxygen 6-8L/minute
- Nothing by mouth: administer IV fluids

Intra-Abdominal Injury (cont.)

Initial Treatment

- Hemoglobin, cross-match, clotting tests
- Assess for blood transfusion
- Administer antibiotics
- Pain management
- Prepare for transfer if laparotomy not available

Uterine perforation

Suspect a uterine perforation if the woman has a history of unsafe abortion and any of the following symptoms:

- a fast heart rate (110 beats per minute or more)
- falling blood pressure (diastolic less than 60)
- excessive bleeding
- excessive pain
- infection

Uterine perforation

- Rarely, uterine perforation can occur during a manual vacuum aspiration (MVA)

During an MVA procedure, suspect a perforation if any of the following occur:

- Sudden increase in bleeding during the procedure.
- An instrument penetrates beyond the expected size of the uterus.
- The syringe vacuum decreases with the cannula well inside the uterine cavity.
- The woman continues to bleed excessively after the uterine cavity is empty.
- The woman's pain increases suddenly
- Fat or bowel is found in tissue from the uterus

- Management depends on whether the evacuation is complete when the perforation is discovered and on the health-care facility's capabilities.

If a perforation is found and the **evacuation** is **complete**:

- Begin IV fluids and antibiotics.
- Give ergometrine (0.2mg IM); repeat as needed up to three doses.
- Monitor closely for two hours; check vital signs and blood loss frequently;
- If the patient remains stable and bleeding slows, give ergometrine PO or im , 8 hrly.
- If condition gets worse, bleeding continues, Laparotomy/ Laparoscopy.

If a perforation is found and the evacuation is **not complete**:

- Begin IV fluids and antibiotics
- check the woman's hematocrit
- Make arrangements for blood transfusion or plasma volume expander, if indicated.
- **Complete the evacuation** under direct visual control (laparoscopy or laparotomy) to assess damage and injury to abdominal organs, such as the bowel
- if direct visual examination is not available, refer the patient to a higher-level hospital.
- Depending on the finding, **Repair** or **Hysterectomy** is done.
-

POST ABORTION CARE

- Is global strategy designed to reduce maternal mortality&morbidity occurring as a result of unsafely induced &spontaneous abortions.Has **five components** namely..
- 1.Emergrncy Tx of incomplete abortion \$potentially life threatening complications.
- 2.Pos abortion family planning counselling&services.

- 3. Links b/n post abortion emergency services & the reproductive health care system.
- 4. Community service provider partnership.
- 5. Counselling

THANK YOU !